Coulomb’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electric force to charge magnitude and separation.

SUCCESS CRITERIA - I can...

1. Identify charge signs before predicting direction.
2. Use inverse-square scaling.
3. Calculate force with charge in coulombs and distance in meters.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A shallow-water model uses $v=\sqrt{(gh)}$. With $g=10$, $h=30$ m and travel distance 5 km, find speed and travel time. State a model limitation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = k|q_1q_2|/r^2$; $k = 9.0\times10^9$ N·m²/C².
r is separation. Signs determine attraction/repulsion; the equation gives magnitude.

Worked reasoning: Doubling the separation between two fixed charges makes Coulomb-force magnitude one quarter as large. The charge signs determine attraction or repulsion separately.

Your turn: Sketch both separations for two positive charges and label the force directions.



Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two opposite charges are moved twice as far apart. Draw their force directions and compare the new force with the original.



r

\+ -

Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

Two small charged spheres carry +2.0 μC and -3.0 μC and are 0.30 m apart. Use $k = 9.0 \times 10^9$ N·m²/C² and 1 μC = 10^{-6} C. Label the model, calculate force magnitude and draw the force on each sphere.



r = ?

Charge 1 Charge 2

Marzano check - circle: ① ② ③

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Continue down each column.

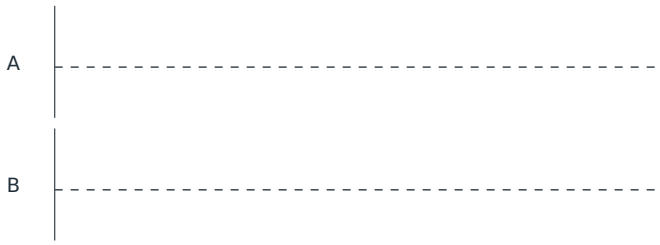
05 Explain your reasoning

One charge has twice the magnitude of the other. Does it exert twice the force on the smaller charge? Explain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 600 Hz tone lasts 0.4 s. Find number of cycles and period.
Label a wave sketch to support the calculation or comparison.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Both charge magnitudes double and distance triples. Express the new force as a fraction of the old.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two small spheres carry +1.0 μC and +4.0 μC and are 0.20 m apart. Use $k = 9.0 \times 10^9$ N·m²/C². Calculate the force magnitude and draw both force arrows. Then triple the separation and recalculate.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3